CRB memory hierarchy: results brief

Phoenix Reference Server, 128 Neoverse V3AE at 2.8 GHz, two CMN S3 meshes. Measured 2026-10-30 to 2026-11-02 on CPUs 0 to 119. Full report with methods: crb-memory-hierarchy.html. Brief generated 2026-11-02.

Levels, one core and whole chip

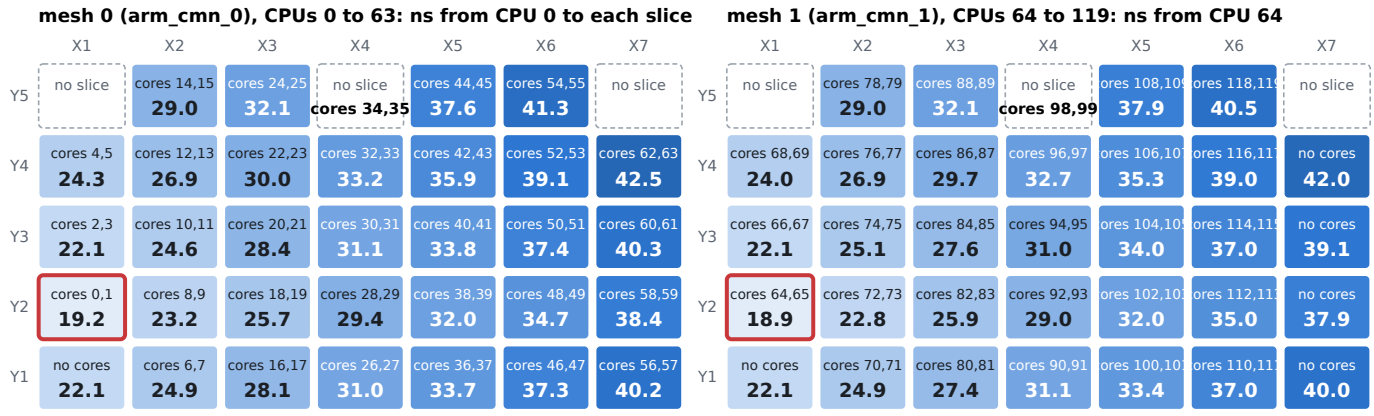
LEVEL	SIZE	LOAD-TO-USE LATENCY	READ, 1 CORE	WRITE, 1 CORE	READ, WHOLE CHIP	WRITE, WHOLE CHIP
L1 data, private, 4-way	64 KiB	1.43 ns (4 cycles)	133 GB/s	89.5 GB/s	15.9 TB/s	10.7 TB/s
L2, private, 8-way	2 MiB	3.75 ns (10.5)	85 GB/s	89.5 GB/s	10.2 TB/s	10.7 TB/s
Mesh cache (SLC), 128 slices of 1 MiB, 16-way	128 MiB (64 MiB seen by one core)	19 to 44 ns by slice, 29 avg (85)	67 GB/s	73 GB/s, goes to DRAM	≥ 0.62 TB/s served by slices (1.38 TB/s total at 40 threads)	not cached
DRAM, 10 ch DDR5-8000	320 GiB	148 ns (415)	30 GB/s	63 GB/s	533 GB/s (83 % of 640 peak)	447 GB/s

Latency: one dependent load, random chain, prefetch defeated. Per-core bandwidth: one thread, hand-written ldp/stp loop. Whole chip: best placement on CPUs 0 to 119. Cycles at 2.8 GHz.

Findings

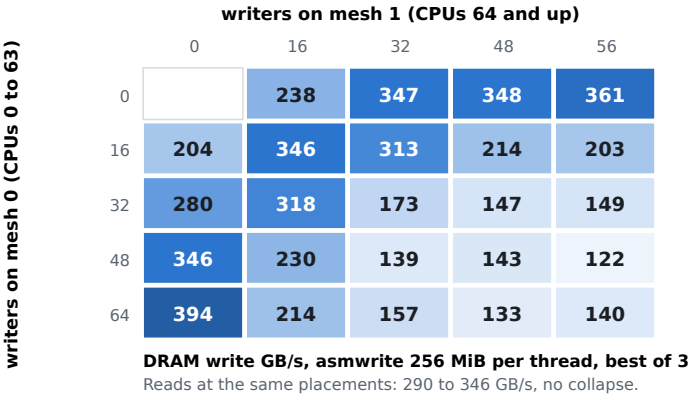
- **Pair port: cores 2k and 2k+1 share one crosspoint port. Two partners together read 78 GB/s from the mesh cache; two non-partners 134 GB/s.**
- One writer's 73 GB/s limit is inside the core: no L1/L2 allocation (0 refills, 0 write-backs), no credit stalls on its crosspoint links, and the same port carries 82 GB/s with the partner core, 229 GB/s with 4 writers [sudo-20261102T211150Z/writepath.csv].
- **Slice distance: nearest slice 18.8 to 22.2 ns, 2.7 ns per crosspoint hop, farthest 32.9 to 44.2 ns. Repeat spread 3 %.**
- Every core's position is known from its latency gradient (map below): CPUs 0 to 63 on mesh 0, 64 to 119 on mesh 1, numbered column by column; pairs 34/35 and 98/99 sit on a crosspoint without slices.
- Every 2 MiB chunk of memory is split half and half between the two meshes by an address hash (the XOR of address bits 6, 8, ... 20 and higher bits), not line by line. A hit is always served on the core's own mesh, by the home slice or by a proxy slice for lines homed on the other mesh, at the same 19 to 44 ns. A line held only on the other mesh costs about 85 ns on its first access, then it is kept locally. So one core's data lives on its own mesh: one buffer sees 64 MiB, not 128.
- Streaming writes largely bypass the mesh cache: 0.013 fills and 0.76 memory requests per line at 8 MiB per thread. The 73 GB/s "mesh write" is mostly the DRAM write path.
- **DRAM write collapses only with heavy writers on both meshes: 64 writers on one mesh 394 GB/s, 32 on each mesh 173 GB/s. Reads do not collapse. Home-node retry and throttle counts fall in the collapsed case; the bottleneck is not located (hypothesis: chip-to-chip links).**
- Concurrent reads from the mesh cache: 32 threads on one mesh share about 869 GB/s (25 to 30 each); 8 threads on the other mesh get 58 to 60 each (3 MiB per thread, one thread per pair, 1372 GB/s total).

Where the cores are, and how far each slice is



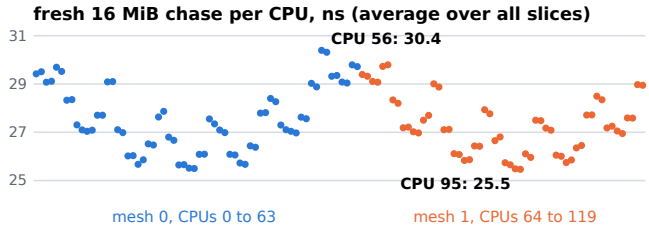
Each cell is one crosspoint with two 1 MiB slices. Top line: the core pair on its port. Number: median of 7 fresh pointer chases of 2048 lines homed there. Red border: the measuring core. Dashed: a crosspoint without slices. Nearest slice 19 ns, about 2.7 ns per hop, far corner 33 to 44 ns.

DRAM write bandwidth against writer placement



DRAM write GB/s, asmwrite 256 MiB per thread, best of 3

Reads at the same placements: 290 to 346 GB/s, no collapse.



Average mesh-cache latency by core: 25.5 to 30.4 ns, partners equal, two sweeps agree to 0.2 ns. The spread is the core's distance to the slices.

What the home-node counters say, per 64 B line streamed

STREAM (THREADS ON MESH 0 CORES)	GB/S	SLICE ACCESSES, OWN MESH	SLICE FILLS	SLICE MISSES	MEMORY REQUESTS, OWN MESH	ACCESSES ON OTHER MESH
asmread 8 MiB, 1 thread	71	0.83	0.83	0.000	0.000	0.000
asmread 8 MiB, 14 threads (112 MiB)	285	0.82	0.82	0.30	0.33	0.027
asmwrite 8 MiB, 1 thread	72	0.76	0.013	0.75	0.76	0.38
asmwrite 1 GiB, 14 threads	238	1.00	0.016	1.00	1.03	0.50

Rates = (2 s run − 0.2 s run) counts over the byte difference, idle background subtracted, every counter at 100 % running time, checker exit status verified. A read takes the line out of the slice and the L2 eviction writes it back (exclusive allocation), hence accesses ≈ fills on reads. Writes mostly miss and go to memory. The other mesh sees nothing for an 8 MiB working set: its half of the lines is served by proxies on the core's own mesh.

Cross-mesh write collapse, per line written

WRITERS	GB/S (LONG RUNS)	WRITE REQUESTS, MESH 0 / 1	WRITE RETRIES, MESH 0 / 1	MEMORY THROTTLING, MESH 0 / 1	MEMORY REQUESTS, LOCAL (MESH 0 + 1) / REMOTE (0 + 1) / ALL
64 on mesh 0	393	1.06 / 0.50	0.058 / 0.000	0.103 / 0.013	1.14 / 0.51 / 1.65
32 on each mesh	141 to 161	0.81 / 0.72	0.013 / 0.012	0.000 / 0.000	1.42 / 0.52 / 1.94
32 on each mesh, reads	328	0.72 / 0.78	0 / 0	0 / 0	1.58 / 0.50 / 2.08

Events per application line, summed as labelled; a remote-homed request is counted at its proxy and at its home, so totals exceed one per line and are not unique writes. Retry and throttle counts fall in the collapsed case; that does not measure home-node utilisation, and the bottleneck is not located. Each event pair comes from its own run, hence the GB/s range.

Open

- Bottleneck of the cross-mesh write collapse: chip-to-chip gateway and memory-controller counters not yet read.
- Slice partition between home lines and proxy lines (configuration register, not perf), and the excess whole-mesh accesses and misses beyond the timed loads, recorded but not attributed.
- Cross-chiplet first access: 85 to 88 ns per load at 16 MiB for lines the other chiplet's slices hold, 32 ns on the second walk (matches Bill's 81 to 88 ns); remote-slice hits and lines still in the builder's L2 not separated.